



Gazi Mohammad Fahim Faiyaz

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AI/ML Engineer specializing in Computer Vision and Generative AI, with industry experience building and deploying real-world AI systems. Skilled in developing end-to-end machine learning solutions and applying deep learning, explainable AI, and LLM-based approaches to practical problems. IEEE-published researcher with a strong focus on creating reliable and scalable AI applications.

CORE TECHNICAL SKILLS

Programming & Tools: Python, C/C++, SQL (MySQL, PostgreSQL), Git, Jupyter Notebook, Tableau

AI/ML & Computer Vision: Machine Learning, Deep Learning, Object Detection, Segmentation, Time-Series Analysis; TensorFlow, PyTorch, Keras, Scikit-Learn, FastAI, OpenCV, Ultralytics

NLP & LLMs: Transformers, Embeddings, RAG, LangChain, LangFlow, Prompt Engineering; VectorDatabase(ChromaDB, AstraDB)

Data Engineering: Selenium, Web Scraping, Automated Data Pipelines, Data Cleaning & Preprocessing

MLOps & Cloud: Docker, GitHub Actions, MLflow, ONNX Runtime, TensorRT, FastAPI, Flask, Gradio, Hugging Face Spaces, Roboflow, CVAT, Supervisely, Encord; AWS (Basic), Azure (Basic)

EXPERIENCE

Computer Vision Engineer

03/2025 – Present

Quantigo AI • Dhaka, Bangladesh

Engineered production-ready annotated image and video datasets for **5+ active computer vision pipelines**, directly reducing model training cycle time and accelerating client deployment readiness.

Validated and deployed **object detection models** across 5 domains including **Street Text Recognition, Microsurgical Scene Analysis, Industrial Defect Annotation, Search & Rescue**, and **Surveillance Event Tagging**, maintaining production quality benchmarks on every release.

Diagnosed annotation inconsistencies and edge-case failures in active **CV pipelines**, improving model robustness and reducing post-deployment error rates across multiple client projects.

PROJECTS

AI / ML & Computer Vision Research

Driver Distraction Detection [GitHub](#) — Real-Time Computer Vision System

2025

Engineered a **YOLOv10 real-time system** classifying **12 distraction behaviors** at **98.1% mAP@50**; slashed inference latency via ONNX Runtime and automated deployment through CI/CD.

Stack: Python, YOLOv10, ONNX Runtime, Flask, Gradio, Docker, GitHub Actions

Mango Ripeness Classification [GitHub](#) — Explainable AI with Xception-LSTM

2025

Designed a novel **hybrid Xception-LSTM architecture** achieving **98–99% accuracy** on 6,000 images; integrated Grad-CAM for transparent predictions. *[Published: IEEE RAAICON 2025]*

Stack: Python, TensorFlow, Keras, Xception, LSTM, Grad-CAM

Brain Tumor Detection [GitHub](#) — YOLOv10 & Hybrid CNN with XAI (BSc Thesis)

2024

Achieved **mAP@50 = 0.964** on MRI tumor detection using YOLOv10 variants; developed hybrid VGG19-LSTM / VGG16-Inception models with Grad-CAM for clinical explainability. *[Published: IEEE RAAICON 2024 & ICCIT 2024]*

Stack: Python, TensorFlow, PyTorch, OpenCV, Ultralytics YOLOv10, Grad-CAM

CS Research Article Classification [GitHub](#) — Multi-Label NLP with SciBERT

2025

Scraped and curated **~30,000 arXiv papers**; fine-tuned SciBERT to **97.4% multi-label accuracy**, exported to ONNX, and deployed as a live Gradio app.

Stack: Python, SciBERT, FastAI, ONNX, Flask, Gradio, Selenium, Docker

Telemetry Anomaly Detector [GitHub](#) — Time-Series Anomaly Detection System

2025

Built a **production-ready 3-layer ensemble** (STL Z-score, Isolation Forest, CUSUM) on NAB server telemetry; CUSUM layer achieves high recall for drift detection while ensemble balances precision-recall tradeoff — served via FastAPI and Streamlit dashboard.

Stack: Python, Scikit-learn, Statsmodels, FastAPI, Streamlit, Docker, Pytest

Generative AI & Multi-Agent Systems

- EmailCraft-AI GitHub — Production Multi-Agent B2B Outreach System2025
Architected a **production-ready multi-agent pipeline** automating personalized B2B outreach by combining RAG retrieval, job-post scraping, persona profiling, and portfolio matching — served via containerized REST API.
Stack: Python, FastAPI, Groq (LLaMA), ChromaDB, Pinecone, SQLite, Docker
- AI Fitness Coach GitHub — Intelligent Conversational Health Assistant2025
Built an **intelligent multi-agent system** with dynamic query routing delivering personalized fitness and nutrition plans, with long-term memory persistence via AstraDB for context-aware interactions.
Stack: Python, Streamlit, LangChain, Groq AI, AstraDB

Data Science & ML Web Applications

- King County House Price Predictor GitHub — End-to-End ML Web Application2025
Developed an **XGBoost regression model** with real-time FastAPI prediction endpoints; containerized via Docker with a full CI/CD pipeline for seamless production delivery.
Stack: Python, Scikit-learn, XGBoost, FastAPI, HTML/CSS, Docker, GitHub Actions
- Bangladeshi Landmarks Recognizer GitHub — Custom Dataset & Image Classifier2025
Independently scraped a **16,741-image landmarks dataset**; trained DenseNet121 to **99.9% accuracy** and deployed as a live Hugging Face Gradio application.
Stack: Python, FastAI, PyTorch, DenseNet121, Gradio, GitHub Pages

PUBLICATIONS

- A Hybrid Deep Learning Approach For Brain Tumor Detection Using XAI with GradCAM.
ICCIT 2024 | Author Position: 1st Author | 27th IEEE Int’l Conf. on Computer and Information Technology (ICCIT), Cox’s Bazar, Bangladesh, 2024, pp. 1235–1240
doi: 10.1109/ICCIT64611.2024.11021963
- Deep Learning for Brain Tumor Detection Leveraging YOLOv10 for Precise Localization.
IEEE RAAICON 2024 | Author Position: 2nd Author | 3rd IEEE Int’l Conf. on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON), Dhaka, Bangladesh, 2024, pp. 207–212
doi: 10.1109/RAAICON64172.2024.10928354
- Deep Learning Approaches for Distracted Driving Detection Using YOLOv10-S and EfficientNet-B0.
ICCIT 2025 | Author Position: 1st Author | 28th IEEE Int’l Conf. on Computer and Information Technology (ICCIT), Cox’s Bazar, Bangladesh, 2025, pp. 1–6
doi: 10.1109/ICCIT68739.2025.11491547
- Tea Leaf Disease Detection Using Explainable AI & the Fusion of Ensemble and Deep Learning.
ICCIT 2025 | Author Position: 3rd Author | 28th IEEE Int’l Conf. on Computer and Information Technology (ICCIT), Cox’s Bazar, Bangladesh, 2025, pp. 3182–3187
doi: 10.1109/ICCIT68739.2025.11491363
- Tea Leaf Disease Detection and Classification Using YOLOv11 and Transfer Learning with Grad-CAM.
ICCIT 2025 | Author Position: 1st Author | 28th IEEE Int’l Conf. on Computer and Information Technology (ICCIT), Cox’s Bazar, Bangladesh, Dec 2025
doi: 10.1109/ICCIT68739.2025.11516092
- Hybrid Xception–LSTM Model for Explainable Mango Ripeness Classification Using Grad-CAM.
IEEE RAAICON 2025 | Author Position: 2nd Author | 4th IEEE Int’l Conf. on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON), MIST, Dhaka, Bangladesh, 2025, pp. 339–344
doi: 10.1109/RAAICON69033.2025.11502078

COURSES & CERTIFICATIONS

- Dokkho MasterCourse (Cohort 7)2025
Data Science Career Program. Selected via competitive assessment; 4 end-to-end capstone projects.
- Generative AI with LangChain & HuggingFace Udemy2025
LLM applications, embeddings, vector DBs, RAG pipelines and agent workflows.
- Complete ML & NLP Bootcamp + MLOps Udemy2024
ML, NLP, feature engineering, Docker deployment and MLOps for real-world AI systems.
- App Development with Flutter Ostad2024
Practical experience building cross-platform mobile apps using Flutter and Dart.

EDUCATION

- B.Sc. in Computer Science2020 – 2025
East Delta University • Chattogram, Bangladesh
CGPA: 3.60 / 4.00 | 5× Dean’s List (GPA 3.5+) | 1× Vice Chancellor’s List (GPA 4.0)
Thesis: Brain Tumor Detection Using YOLOv10 and Hybrid CNN Models with Explainable AI

LANGUAGES

Bangla — Native | English — Fluent (Reading, Writing, Speaking, Listening)